

Project Report

Humidity Regulator

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Abstract: This project creates a smart humidifier using an **ESP microcontroller** and **DHT11 sensor** to automatically regulate indoor humidity for optimal comfort.

Introduction

Maintaining an optimal indoor environment is crucial for both comfort and health. One of the key aspects of indoor environmental quality is **humidity**. Humidity refers to the amount of moisture in the air, and when it falls below the recommended levels, it can cause various health discomforts and damage to the living environment. The goal of this project is to design and implement a **humidity regulation system** that uses smart technology to monitor and adjust the room's humidity levels. This report outlines the development and implementation of a **smart humidifier** system that automatically activates when the room's humidity falls below a set threshold. Using an **ESP microcontroller**, **DHT11 sensors**, and **piezoelectric transducers**, this system is designed to ensure that the humidity remains within the ideal range for human comfort, health, and well-being.

Objective

The primary objective of the project is to create an efficient system that automatically maintains **optimal humidity levels** in a room by monitoring temperature and humidity using sensors. The system will activate when the humidity level drops below a predetermined threshold, and it will stop when the humidity returns to the desired level. This smart regulation ensures comfort, prevents health discomfort, and protects furnishings and electronics from the adverse effects of low humidity.

Components Used in the Humidity Regulation System:

- **ESP Microcontroller:**
 - Controls the system and activates the humidifier based on data from the **DHT11 sensor**.
- **DHT11 Sensor:**
 - Measures **temperature** and **humidity**, sending data to the ESP for system control.

- **Piezoelectric Transducer:**
 - Generates ultrasonic mist to increase room humidity.
- **555 Timer IC:**
 - Controls the timing and frequency of the ultrasonic transducer.
- **MOSFET (TR1):**
 - Switches the ultrasonic transducer on and off based on the timer's signal.
- **Resistors (R1, R2):**
 - Set timing for the **555 timer** and control current flow through the circuit.
- **Capacitors (C1, C2, C3):**
 - Stabilize the timer's frequency and sensor readings.
- **Inductor (L1):**
 - Ensures the circuit operates at the correct frequency for mist generation.
- **Power Supply (12V to 15V DC):**
 - Powers the system components.

Methodology

1 System Setup:

- **Sensor Data Collection:** The **DHT11 sensor** was placed in the room to measure the temperature and humidity continuously.
- **Microcontroller (ESP):** The **ESP microcontroller** collects data from the DHT11 sensor. When the humidity falls below a threshold (set at 50% RH for this system), it sends a signal to activate the **piezoelectric transducer** to generate water vapor.
- **Humidifier Activation:** The humidifier runs until the humidity reaches the desired level. Once the system detects an increase in humidity, the humidifier is automatically turned off.
- **Environmental Conditions Monitored:** Data was gathered in three room conditions:
 - **Closed Room with Fan On**
 - **Air Circulation with Open Room**
 - **Closed Room without Air Circulation**

2 Data Collection:

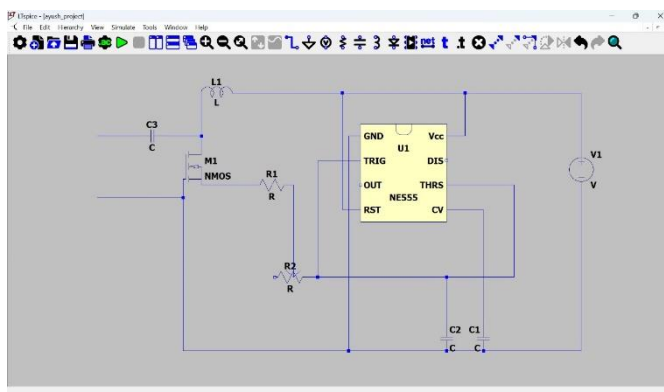
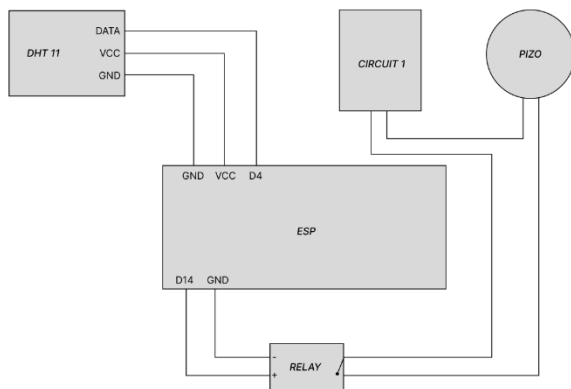
Humidity and temperature readings were taken over a period of 5 hours, with measurements recorded at regular intervals. The data was then used to create

graphs using **Matplotlib** for analysis. These graphs helped to observe the fluctuations in **temperature** and **humidity** levels across different room conditions and to pinpoint when the **humidifier should be activated**.

3 Key Parameters:

- **Threshold Humidity Level:** 50% RH (Relative Humidity)
- **Room Size:** 80 cubic meters
- **Graphing:** Humidity vs. Time and Temperature vs. Time graphs were created to visualize the behavior of the system in different conditions.

4 Circuit Diagram:



Circuit 1

5 Explanation of Components:

1. DHT11 Sensor:

- The **DHT11** is a **temperature and humidity sensor** that measures the current environmental conditions (temperature and humidity). It sends this data to the **ESP microcontroller**, which processes the information and makes

decisions based on the programmed threshold for humidity.

○ Connections:

- **VCC:** Powers the sensor (typically 5V or 3.3V depending on the microcontroller).
- **DATA:** The data pin connected to the **ESP microcontroller**.
- **GND:** Ground connection.

2. ESP Microcontroller:

- The **ESP** (likely ESP8266 or ESP32) is the central control unit for the system. It receives data from the **DHT11 sensor**, processes it, and activates the **relay** to power the **mist-making circuit** when the humidity falls below a set threshold (e.g., 50% RH).
- **Connections:**
 - **VCC:** Provides power (usually 3.3V or 5V).
 - **D4 Pin:** Controls the relay or activates other connected components.
 - **D14 Pin:** A GPIO pin that might be used to control specific functions, such as the relay or triggering certain actions.
 - **GND:** Ground connection.

3. Relay:

- The **relay** is used to switch higher-power components (like the **ultrasonic mist maker**) on and off based on the signal from the **ESP microcontroller**. The microcontroller can only handle low-power signals, so the relay acts as a switch to control the higher-power circuits, such as the **mist maker**.
- **Connections:**
 - **Relay Trigger (D4):** The relay is activated through this pin when the **humidity threshold** is crossed.
 - **Common (COM), Normally Open (NO), and Normally Closed (NC):** These relay terminals control the flow of current to the **mist-making circuit**.

4. Piezoelectric Transducer (PIZO):

- The **piezoelectric transducer** is used to generate the ultrasonic waves that turn water into a fine mist. This mist increases the humidity in the room. It works by vibrating at a high frequency when driven by a signal, usually provided by a **555 timer circuit**.
- **Connection:**
 - It is controlled by the **mist maker circuit**, where the power is supplied through the **relay** to the transducer.

5. 555 Timer IC:

- The **555 timer** is used in the **astable mode** to generate a continuous square wave that drives the **piezoelectric transducer**. It controls the frequency at which the transducer vibrates, producing ultrasonic waves.

- **Components connected to 555 timer:**
 - **C1, C2, and C3 (Capacitors):** These capacitors help set the frequency of the timer and stabilize the signal. Capacitors **C1** and **C2** are involved in timing and frequency control, while **C3** is used for smoothing out noise and ensuring stable operation.
 - **R1 and R2 (Resistors):** The resistors work with the capacitors to determine the timing characteristics and frequency of the output signal.
 - **L1 (Inductor):** It is used to help fine-tune the oscillation frequency, creating the necessary resonance needed for ultrasonic mist generation.
- 6. **Transistor (TR1):**
 - The **transistor** acts as a switch for the **piezoelectric transducer** and is controlled by the **555 timer** output. When the **555 timer** generates a signal, the transistor allows the power to flow to the transducer, causing it to vibrate and generate mist.
 - **Connections:**
 - **Base (B):** Connected to the **555 timer** through the resistor network.
 - **Collector (C):** Connected to the ground side of the **piezoelectric transducer**.
 - **Emitter (E):** Ground connection.
- 7. **Inductor (L1):**
 - The **inductor** (L1) works with the capacitors to filter and stabilize the frequency generated by the **555 timer**. It helps in tuning the circuit to the correct oscillation frequency, which is required to drive the **piezoelectric transducer** efficiently.
- 8. **Power Supply (12V to 15V):**
 - The **mist-making circuit** requires a higher voltage (12V to 15V) for the **piezoelectric transducer** and the **555 timer IC**. This voltage powers both the **mist-making circuit** and the **ESP microcontroller** when it is connected to the system.
 - **Connection:**
 - The power supply is connected to the **mist maker circuit** as well as the **relay** and **ESP**.

Health Discomfort Due to Low Humidity

Low humidity can have a significant impact on human health and comfort. Some of the common discomforts associated with low indoor humidity levels include:

1. Skin Dryness and Irritation

- **Dry Skin:** Skin becomes dry and cracked due to moisture loss.
- **Chapped Lips:** Lip skin becomes cracked and painful.

2. Respiratory Issues

- **Dry Nasal Passages:** Causes congestion, irritation, and nosebleeds.
- **Increased Risk of Infections:** Dry mucous membranes are more prone to infections.
- **Sore Throat and Coughing:** Low humidity causes throat irritation, leading to a persistent cough.

3. Eye Discomfort

- **Dry Eyes:** Irritation and discomfort in the eyes, often leading to redness and blurred vision.
- **Increased Risk of Infections:** Dry eyes are more susceptible to infection.

4. Dehydration

- **Fluid Loss:** Prolonged exposure to low humidity can lead to increased fluid loss from the skin and respiratory tract.
- **Fatigue:** Dehydration often causes fatigue, dizziness, and discomfort.

5. Impact on Sleep Quality

- **Disrupted Sleep:** Dry air can make breathing difficult, causing snoring and disturbed sleep.
- **Increased Risk of Sleep Apnea:** Dryness in the throat and nasal passages may aggravate sleep apnea.

Results and Data Analysis

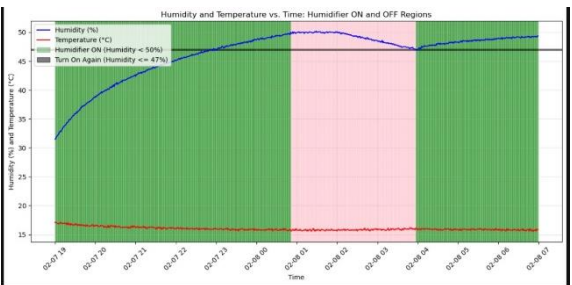
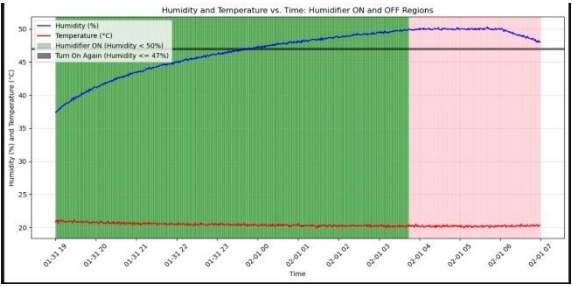
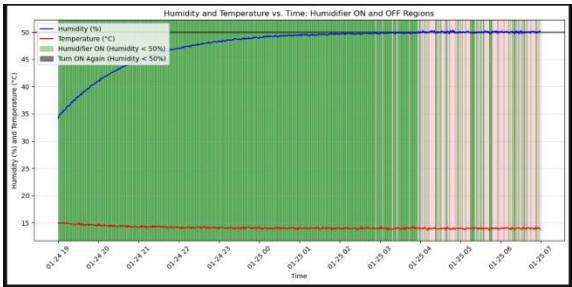
Graphical Data Analysis:

1. Humidity vs. Time: The graph shows how the room's humidity fluctuates under the three conditions. Significant drops in humidity were observed, particularly in the **closed room with fan on** condition, indicating the need for more frequent humidifier activation. The **open room with air circulation** condition showed more stable humidity levels, reducing the activation time.

2. Temperature vs. Time: Temperature remained fairly constant across all three conditions, with minor variations. This indicates that **temperature** does not have as strong an impact on humidity as initially anticipated, but it still provides valuable data for understanding overall room comfort.

Humidifier Activation:

- **Closed Room with Fan On:** The most frequent activation of the humidifier due to significant drops in humidity.
- **Air Circulation in Open Room:** Less frequent activation due to stable airflow, which helps maintain constant humidity.
- **Closed Room Without Air Circulation:** Occasional activation, as the room maintains a relatively stable humidity level.



The objective of our project was to maintain indoor relative humidity (RH) at 50%, an ideal level for health and comfort. Initially, the system was set to activate the humidifier exactly at 50% RH. However, during real-time testing, we observed that even minor RH fluctuations caused the humidifier to switch on and off frequently. This led to two major issues: excessive wear and tear on the humidifier and sensor, and unstable humidity levels. To address this, we introduced a hysteresis control strategy. Instead of reacting exactly at 50%, the humidifier now turns ON when RH drops to 47% and turns OFF once it reaches 50%. This approach significantly reduced switching frequency, stabilized humidity control, extended the equipment's lifespan, and optimized energy usage. In our experiments, we also compared humidifier performance in different environments. In a closed room, the humidifier quickly reached and maintained the target RH due to minimal air exchange and moisture loss. In contrast, an open ventilated room made it harder to maintain humidity because the constant airflow removed moisture from the air. This study emphasized the importance of minimizing ventilation and sealing the environment to ensure efficient and effective humidification.

Key Takeaways:

- The **optimal humidity threshold** was set at **50% RH**, ensuring comfort without wasting energy.
- **Health issues** due to low humidity, such as skin dryness, respiratory discomfort, and dehydration, were effectively mitigated through the regulation of humidity.
- The system demonstrated **adaptive control**, adjusting to different environmental conditions to maintain consistent indoor humidity.

Future Recommendations

- **Smart Home Integration:** Future versions could integrate with home automation systems, allowing users to control the humidifier remotely via mobile apps.
- **Energy Efficiency:** Further improvements can focus on optimizing the piezoelectric system to consume less power.
- **Air Quality Sensors:** Additional sensors, such as **PM2.5** and **CO2** sensors, could be integrated to monitor air quality and make the system even more effective at improving indoor environments.

Competitive Landscape Overview of the Market

The humidifier market is growing, with various products available for home and office use. However, most traditional humidifiers are manual and do not adjust based on real-time environmental conditions. Here's how our system compares:

Feature	Traditional Humidifiers	Our Humidity Regulation System
Automatic Operation	No	Yes
Real-Time Monitoring	No	Yes
Integration with Smart Systems	No	Future integration potential
Health Benefits	Limited	Enhanced comfort and health
Energy Efficiency	Varies	Optimized for minimal usage

Key Advantages:

- **Real-time Humidity Control:** Unlike traditional humidifiers that run continuously, our system adjusts based on live data, ensuring efficient energy use.
- **Health-Focused:** The system helps prevent health discomforts such as dry skin, sore throats, and respiratory issues, which are not directly addressed by traditional humidifiers.
- **Smart System Potential:** Future upgrades can integrate with smart home systems, providing users with more control and remote access.

Conflict of Interest

The authors declare no conflict of interest regarding the development, testing, and publication of this project. All components and systems discussed in this report are proprietary and developed by our team at Indian Institute of Technology Delhi.

Conclusions

The project successfully designed and implemented a **humidity regulation system** that provides automatic humidification to maintain comfort and health in indoor environments. The results from our data indicate that the system works effectively in different room conditions, with the **humidifier activating when humidity falls below 50% RH** and deactivating once optimal humidity levels are restored.

References

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